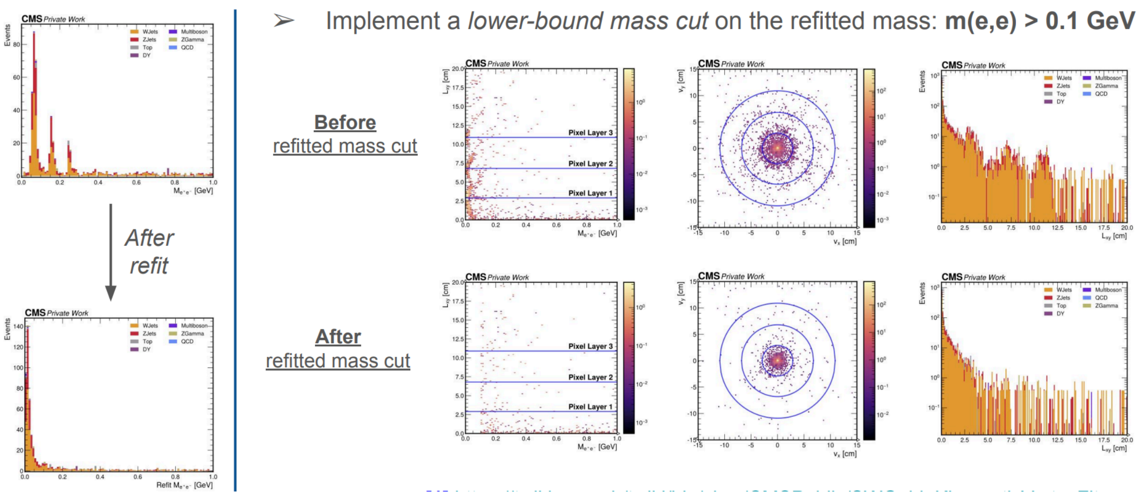
Conversion Background

Conversion backgrounds in Displaced (Di-)Electron Final State

In searches with displaced electrons in the final state, i.e. displaced di-electron secondary vertex (SV), photon conversions can show up as background. While conversion backgrounds can be suppressed with electron ID and `conversionVeto` [1][2], one way to further supress the conversion background is to make a lower-bound SV mass cut. This is based on the features of conversion that its reconstructed mass of the di-electrons should be near zero as they decay from photons.

Di-electron Secondary Vertex and Cut on the Refitted Mass

Once the dielectron SV is formed with the vertex fitting algorithm (i.e. Kalman Vertex Fitter [3]), one can use the lower-bound cut on the di-electron SV mass. For this, re-computation of SV mass based on SV track refit is recommended. For example, Kinematic Vertex Fit [4] module can refit the electron tracks based on SV constraint and re-compute its mass. Without re-computing the mass with refit, multiple peaks corresponding to the conversion are observed from SV low mass region. After the refit, the multiple mass peaks converge to a single peak near zero, on which one can make a cut to reject the conversion backgrounds. The effect of the lower-bound mass cut in its conversion rejection power can be seen from SV mass vs. Lxy plot.



More detailed information can be seen from from this talk.

References

[1] https://twiki.cern.ch/twiki/bin/viewauth/CMS/ConversionTools

[2] https://github.com/cms-sw/cmssw/blob/master/PhysicsTools/PatAlgos/plugins/PATElectronProducer.cc#L748-L749

[3] https://twiki.cern.ch/twiki/bin/view/CMSPublic/SWGuideKalmanVertexFitter

[4] https://twiki.cern.ch/twiki/bin/view/CMSPublic/SWGuideKinematicVertexFit

Contributors

Kyungmin Park, on behalf of the Inelastic Dark Matter (with Soft Displaced Electrons) analysis team